

Malware via USB Device response

Malware · **High** severity · SOC IR Playbook

Incident type	Physical Media-Based Malware Infection
Severity	High
Priority	High if lateral movement or sensitive data is involved
Detection sources	EDR, antivirus/antimalware, SIEM, user report, USB monitoring tools

Scenario

Malicious software is introduced into the environment through an infected USB storage device. This may include autorun malware, ransomware, keyloggers or tools used to establish persistence or exfiltrate data.

MITRE ATT&CK

T1059, T1091, T1200

Preparation

Disable USB autorun Group Policy settings or endpoint hardening Implement USB control software Allow only authorised devices; log USB insertions Enforce endpoint protection EDR with removable media protection and behavioural detection Educate users Train staff not to plug in unknown USB drives Log USB usage Enable audit policies for removable media activity

Detection & Analysis

Detect malware activity Alert from antivirus, EDR or SIEM on process execution from USB Identify USB event Review logs for usb-storage, DevicePlugEvent or Removable Storage Device events Analyse file origin Determine whether execution started from USB drive (e.g., drive D:\ or E:) Collect indicators Hashes, filenames, execution chain, affected systems

Containment

Isolate infected system Disconnect network and USB ports to prevent spread Remove USB device Preserve for forensic investigation if necessary Block malicious file hashes In EDR or AV systems across the organisation Identify other exposed systems Scan for similar infections or shared lateral movement paths

Eradication

Remove malware Use EDR or AV tools to clean the infected files and processes Delete suspicious files From temporary folders, startup directories or root of USB drive Remove persistence mechanisms Check registry run keys, scheduled tasks, services Perform full malware scan On the infected host and nearby systems

Recovery

Restore system From clean backup if necessary Reinstate connectivity After confirming the host is clean Enable stricter USB policies Allow only whitelisted devices or disable USB entirely in high-risk environments Document the root cause Device origin, user involved, type of malware, system impact

Lessons Learned & Reporting

Conduct awareness training Reinforce security policy on device usage Update USB policy Improve endpoint controls and documentation procedures Share findings with security team Review detection gaps, response time and behavioural indicators Document incident report Include all actions taken, findings and recommendations

Tools typically involved

Endpoint Detection and Response (e.g., CrowdStrike, Cortex XDR); USB control solutions (e.g., DeviceLock, Endpoint Protector, Microsoft Intune policies); Antivirus software (e.g., Windows Defender, Bitdefender, Kaspersky); SIEM for USB and file execution logging; Windows Event Logs (Event ID 2003, 2102 for device

insertion)

Success metrics (SLA targets)

Detection Time <10 minutes after USB malware execution Isolation Time <15 minutes after confirmation Malware Removal Time <1 hour (if no system rebuild required) USB Policy Enforcement 100% of endpoints have policy applied User Awareness Rate ? 90% of users aware of USB risks post-training